Influencer Selection
Network Optimization

Abstract

Streamlit Dashboard app

This project studies the optimal selection of social media influencers for advertising campaigns under constraints using 400K node Twitter network data. We model the Twitter follower data as a directed graph and identify candidate influencers based on PageRank. To keep the computational costs at a reasonable level, a subset of users is randomly sampled to represent the target audience, and influencer candidates are restricted to those appearing within this sample. Since real demographic information is unavailable, we augment the network with synthetic user attributes, including age, gender, region, and primary interest, to simulate a realistic marketing context. The problem is formulated as an integer linear program and solved using Gurobi, with the objective of maximizing audience reach under constraints on budget, influencer reliability, demographic balance, frequency capping, and coverage.

I. Introduction

Influencer marketing has become a major component of digital advertising, with firms spending billions of dollars annually on partnerships with social media creators. However, selecting influencers based solely on follower counts often leads to inefficient campaigns due to substantial audience overlap. When multiple influencers share the same followers, advertisers may pay repeatedly to reach the same users, reducing return on advertising spend. This challenge is particularly pronounced on platforms such as Twitter (X), where directed follower relationships create asymmetric influence patterns and complex network structures. The central question addressed in this project is an optimization problem: given a fixed budget and practical constraints, which influencers should be selected to maximize unique audience reach?

We study this question through the **Social Network Reach Optimization Problem**, a constrained variant of the Maximum Coverage problem. The Twitter network is modeled as a directed graph, where nodes represent users and edges represent follower relationships. Binary decision variables determine whether an influencer is selected and whether a user is reached. The objective is to maximize total unique coverage while ensuring each user is counted at most once. In addition to a budget constraint, the model incorporates several real-world considerations, including minimum average influencer reliability, demographic balance requirements to promote inclusive marketing, and frequency capping to prevent audience overexposure.

Empirically, we use the Higgs-Twitter dataset, augmented with synthetic demographic and reliability attributes. The Higgs dataset contains 456,626 nodes with 14,855,842 edges. Influencer candidates are identified using PageRank, and sampling is employed to manage computational complexity. Overall, this project demonstrates how rigorous optimization modeling and algorithmic comparisons can significantly outperform naive influencer selection strategies, illustrating the practical value of optimization methods in real-world network-based decision problems.

II. Methodology

In this project, we apply two complementary methods to address the optimization problem. First, we formulate and solve the problem using the **Integer Linear Programming (ILP)** model, which serves as the overarching modeling and optimization method. To address the lack of features in the initial follower dataset and make the optimization problem realistic for real-world advertising, we generated **synthetic features** for each node to enable targeted constraints. We assigned demographic data, such as age, gender, and interests, to every user in the graph. During the modeling stage, we adopt a **feature-mapping strategy** that converts each generated attribute into dictionary objects, enabling efficient lookup and association with individual nodes in the network. To identify candidate influencers, we identified the **top 1% most connected nodes** instead of random selection to mirror the structure of actual social networks. We further enriched the model by generating operational metrics for these candidates, including hiring costs and 'reliability scores,' which simulate the real-world risk of an influencer failing to post.

We define a Integer Linear Programming problem as below:

1. Sets & Indices
    - I: Set of candidate influencers
    - U: Set of target audience users
    - F_u: Set of influencers followed by user u
2. Parameters
    - c_i: Cost to hire influencer i
    - r_i: Reliability score of influencer i
    - B: Total Budget
    - R_{min}: Minimum average reliability score required
    - P_{female}: Target proportion of female audience
    - g_u: Gender coefficient for user u (1 if Female; 0 otherwise)
    - K: Frequency cap for spam prevention
3. Decision Variables: $\{x_i\}$ – Binary variable indicating whether we select node i as influencer. $y_u \in \{0,1\}$ means whether user u is reached.
4. Objective: Maximize total nodes reached by influencer selection.

$$\max \sum_{u \in U} y_u$$

5. Constraints:
    a. Budget: total hiring cost must be less than or equal to budget.
        i. $$\sum_{i \in I} c_i x_i \leq B$$
    b. Reliability: average reliability score of the hired influencer must be greater than a certain threshold.
        i. $$\sum_{i \in I} r_i x_i \ \geq \ R_{\min} \sum_{i \in I} x_i$$
        This constraint ensures that the average reliability of all hired influencers is at least the required minimum level. In other words, when we hire multiple influencers, their combined reliability must meet the quality standard Rmin.
        ii. $$\sum_{i \in I} x_i \geq 1$$
        This constraint guarantees that at least one influencer is selected, so the campaign is not empty.
    c. Demographic Targeting: we want our reached audiences to be at least x% female.
        i. $$\sum_{u \in U} (g_u - P_{female}) y_u \geq 0$$
    d. Spam Prevention: maximum times a user will see the ad promotion from an influencer.
        i. $$\sum_{i \in F_u} x_i \leq K \quad \forall u \in U$$
    e. Coverage Constraint: $y_u = 1$ if user u is counted as reached. This constraint ensures that we can only serve users who follow at least one hired influencer.
        i. $$y_u \leq \sum_{i \in F_u} x_i \quad \forall u \in U$$

We utilized the Gurobi solver to tackle this optimization challenge. A key step in our approach was translating the graph's complex connectivity into linear programming constraints by

constructing a **bidirectional mapping** (influencer-to-user and user-to-influencer). This allowed us to feed the network structure directly into the solver alongside our demographic features.
We considered using a Genetic Algorithm, which applies an evolutionary approach to iteratively improve the solution. However, we chose Gurobi to prioritize the mathematical guarantee of an **optimal solution**, rather than settling for the approximate results often produced by heuristic methods.

In the second phase, we deployed our model to create a product **Streamlit Dashboard app**. For our deployment, we used **Snowball sampling**, which reduces the graph to connected clusters, in order to reduce our graph size to 20k nodes. In the dashboard, the user can customize their business requirements (constraints) and the model will return the optimal influencers to hire to meet those requirements. These constraints include Total Budget, Min Avg Reliability Score, Min Female Audience (%), and Max Frequency (Spam Cap). The user can also change the sample size depending on what they need. Upon optimizing, our dashboard informs the user of the optimal influencers to hire, the reach, total cost, and average reliability.

III. Results & Discussion

We benchmark our final models on the fully connected graph under the following constraints:

Budget = 25000
Minimum Average Reliability = 0.2
Target Female Audience Percentage = 0.1
Spam Prevention Max View Count = 20

Our Constrained Model hired 31 influencers at $24,929 total and reached 65,655 people. As a random selection baseline, we hired 31 influencers at $101,423 total and reached 28,382 people. (Note: the baseline model is unconstrained and illustrates what happens without optimization) Compared to the baseline, our optimized solution achieves over a 130% increase in audience reach while costing only 25% of the baseline cost.

The combination of our optimization model and dashboard can be extremely useful to businesses, especially in the advertising and campaign space. By maximizing unique reach and limiting repeated exposures instead of simply maximizing follower count, we allow businesses to reduce spending on redundant audiences. The reliability constraint provides greater confidence that campaigns will be executed as expected, reducing risk of failed campaigns. Finally, the demographic constraints enable businesses to focus on their desired market segments, maximizing both influence and overall campaign impact.